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RESEARCH ARTICLE



RBC-AD: conformal anomaly detection with explicit false-alarm control for the Tennessee Eastman Process

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ABSTRACT

Fault detection in the Tennessee Eastman Process (TEP) is challenged by transient dynamics, post-injection regime transitions, and score drift that destabilize fixed-threshold detectors and increase false alarms. In the proposed Risk-Budgeted Conformal Anomaly Detection (RBC-AD) framework, a calibration-based anomaly detector integrates a causal forecasting backbone, frequency-aware scoring, and conformal decisioning. The forecaster is a dilated temporal convolutional network with causal self-attention, trained on fault-free runs to predict one-step-ahead Gaussian parameters by minimizing negative log-likelihood. Detection then scores each window using a fused nonconformity measure that combines the window-averaged predictive log-likelihood with an FFT-based frequency deviation term, using robust normalization fitted on a held-out calibration split and tuning the fusion weight under leakage-controlled splits. Conformal quantiles at miscoverage level $\alpha = 0.05$ define normal, uncertain, and anomaly states with finite-sample correction. Run-level partitioning over 500 runs per split and post-injection sampling (3,000 normal windows, 8,000 fault windows) achieves ROC-AUC = 1.000 and PR-AUC = 1.000 under the post-injection-only protocol; full-stream results provide the deployment-relevant assessment. Static conformal thresholds ($\theta_u = 0.1977$, $\theta_a = 0.3033$) yield F1 = 0.9989 and F1 = 0.9999. Deployment selects an uncertainty threshold to satisfy an empirical normal alarm-rate target of 0.05. Full-stream evaluation reports uncertainty rate and detection delay under k -consecutive alarm logic, with optional drift-triggered recalibration using normal-stream scores.

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Tennessee Eastman Process; anomaly detection; conformal prediction; FAR control; temporal convolutional network

1. Introduction

Chemical and industrial plants rely on dense instrumentation and automated control loops that connect process sensors, manipulated variables, and supervisory controllers within continuous monitoring and actuation infrastructures (Colosimo et al., 2024; Vermesan et al., 2022). These infrastructures generate high-frequency measurements of plant state and support control actions needed to maintain product quality, safety constraints, and throughput under time-varying operating loads (Chen et al., 2020; Li et al., 2022). As automation shifts toward distributed and edge-enabled architectures, fault detection becomes a core requirement for streaming process telemetry (Chiang et al., 2022; Liu et al., 2025). Process-monitoring research has moved from single-model statistical schemes toward hybrid fusion paradigms that combine data-driven learning, process knowledge, uncertainty handling, and multi-source information for more reliable detection in complex industrial systems (Ali et al., 2025). The same review identifies calibration, uncertainty quantification, fusion architecture design, and deployment governance as unresolved requirements for advanced process monitoring. These requirements are directly connected to false-alarm control and stream-level validation in industrial anomaly detection. Sensor noise, cross-variable dependence,

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controller compensation, and scheduled setpoint changes produce multivariate dynamics in which abnormal behaviour can overlap with admissible transient responses (Ren et al., 2023; Wu et al., 2021; Yerimah et al., 2022). Fault signatures may appear gradually, change after injection, or propagate indirectly through coupled variables, which constrains fixed-threshold detectors and pointwise alarm rules (Ahmed et al., 2025; Chen et al., 2020). Missed detections can lead to unsafe operation and equipment damage, whereas excessive false alarms increase operator workload, create alarm fatigue, and delay corrective action (Jalililab et al., 2023). Industrial fault detection therefore requires temporal modelling, stable behaviour across regime transitions, calibrated uncertainty handling, and explicit control of false-alarm behaviour during continuous-stream deployment (Farzaneh & Simeone, 2025; Tajmouati et al., 2024).

1.1. Motivation and problem statement

The Tennessee Eastman Process (TEP) is a benchmark chemical process widely used to evaluate fault detection and process monitoring methods (Downs & Vogel, 1993; Reinartz et al., 2021). It represents a closed-loop industrial plant with interconnected units, including a reactor, condenser, vapour-liquid separator, stripper, compressor, recycle loop, feed streams, and product streams (Downs & Vogel, 1993; Ricker, 1996). The process contains strong interactions between measured process variables and manipulated control variables. A fault introduced in one unit may therefore propagate through several units before it becomes clearly observable (Ricker, 1996; Tian & Hoo, 2005). This structure makes TEP suitable for evaluating anomaly detection under nonlinear dynamics, controller compensation, delayed fault propagation, and transient operating behaviour (Reinartz et al., 2021; Tian & Hoo, 2005).

TEP formalises recurring difficulties in industrial fault monitoring, including high-dimensional sensing, cross-variable coupling, delayed fault effects, and regime transitions (Andersen et al., 2022; Reinartz et al., 2021). Measured departures from baseline behaviour often occur together with control-driven or load-driven transients, which makes separation between abnormal dynamics and nominal process responses difficult (Heo & Lee, 2019; Xu et al., 2022). Recent integrated machine-learning monitoring for physio-chemical process safety combines distributed correlation modelling, locality-preserving projection, and entropy-based information measures, with validation on distillation and TEP benchmarks (Ali et al., 2024). Multi-model fusion monitoring has also been proposed to combine DCCA, autoencoder representations, and reconstruction-based contribution analysis for complex physio-chemical systems (Ali et al., 2025). These studies indicate that current monitoring research increasingly uses fused learning structures for nonlinear, dynamic, and safety-critical processes.

These developments leave several deployment-level limitations unresolved. Many deep-learning and machine-learning detectors still rely on fixed thresholds, implicit test-dependent tuning, or sampled-window evaluation protocols that do not characterise continuous-stream alarm behaviour (Bakdi & Kouadri, 2018; Zhang et al., 2024). These choices can obscure false-alarm behaviour, reduce reproducibility, and inflate reliability estimates under deployment-relevant conditions (Zhou et al., 2024). Benchmark results therefore need to be interpreted with respect to threshold selection, calibration-data separation, and continuous alarm formation rather than isolated classification scores. In operating plants, alarm flooding and threshold instability reduce operator trust, particularly under regime shifts and distributional change (Huang et al., 2022; Yang et al., 2022). Practical monitoring systems require explicit false-alarm control, calibrated uncertainty states, and robustness to post-injection regime transitions without dependence on test-set tuning (Chakraborty & Finkelstein, 2024; Yang & Kuchibhotla, 2025). Existing approaches address these requirements only partially, leaving a gap between benchmark performance and deployable fault detection.

1.2. Overview of Risk-Budgeted Conformal Anomaly Detection (RBC-AD)

The RBC-AD framework combines probabilistic forecasting with calibrated scoring to support leakage-free fault detection under explicit alarm constraints in multivariate process streams (see Figure 1). One-step-ahead dynamics are modelled using a causal dilated temporal convolutional network with residual blocks that outputs Gaussian parameters, enabling window-level likelihood scoring through negative log-likelihood (NLL). A frequency feature bank is fitted on calibration-normal windows to compute spectral

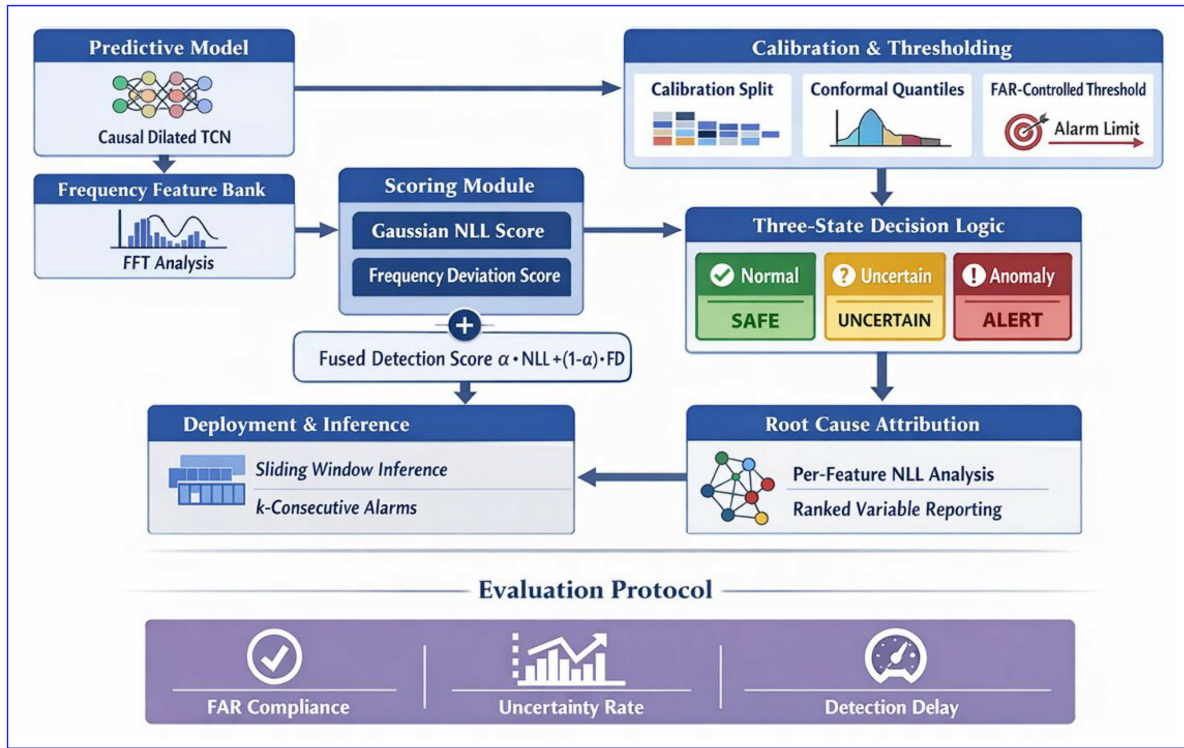


Figure 1. Overview of the proposed RBC-AD framework for leakage-controlled fault detection under false-alarm constraints.

deviation scores from windowed signals. These features are robustly normalised using median and MAD statistics estimated on the calibration-normal split. A fused detection score is defined as a weighted combination of normalised NLL and normalised frequency deviation, parameterised by a mixing weight λ selected on a dedicated tuning split to avoid calibration leakage. Conformal quantiles computed on calibration-normal windows determine an uncertainty threshold and an anomaly threshold, producing a three-state decision space (normal, uncertain, anomaly) for risk-aware monitoring.

Deployment behaviour is specified through a false-alarm-rate (FAR) operating point derived from a normal-only stream. A FAR-controlled threshold is selected on calibration-normal trajectories using stream scoring with a fixed detection stride and optional smoothing, preserving separation between threshold selection and final reporting. During inference, sliding windows generate score sequences that are aggregated using k -consecutive alarm logic to suppress spurious triggers and enable delay trade-offs. Root-cause attribution uses per-feature NLL contributions computed during inference. These contributions are then contextualised using a sparse dependency graph estimated from calibration-normal contribution statistics. This enables ranked variable reporting while accounting for correlation structure among process variables, rather than treating feature scores independently. The evaluation protocol reports sampled-window discrimination and full-stream behaviour, including uncertainty rates, FAR compliance, and detection delay under post-injection trajectories.

1.3. Contributions of this study

This study provides methodological and evaluation contributions for fault detection in the Tennessee Eastman Process (TEP) under operational false-alarm constraints. This paper is classified as an original research article in applied process monitoring and industrial anomaly detection. Methodologically, it presents a calibration-based unsupervised anomaly detection framework for multivariate time-series process monitoring, with emphasis on false-alarm control, uncertainty-aware decisioning, and deployment-oriented evaluation. The contributions are as follows:

- A risk-budgeted detection pipeline is presented that couples a causal dilated TCN forecaster with an auxiliary FFT-based frequency deviation module. Window-level Gaussian NLL and frequency scores are robustly normalised on calibration-normal data and fused into a single detection score using a tuned mixing weight, supporting consistent scoring across process variables and operating regimes.
- A conformal calibration and decision layer is formulated to map the fused score to a three-state output (normal, uncertain, anomaly) with finite-sample correction at a specified significance level. A FAR-controlled deployment threshold is selected on a normal-only calibration stream to meet a target alarm rate, separating threshold selection from final test reporting and aligning the decision rule with alarm-management requirements.
- A leakage-controlled run-level protocol is specified for fitting, validation, tuning, and calibration, with standardisation fitted on training-normal runs only. The evaluation reports sampled-window discrimination metrics, including ROC-AUC and PR-AUC, and full-stream behaviour, including precision, recall, uncertainty rates, and detection delay under k -consecutive logic, enabling reproducible assessment under stream-like conditions.

1.4. Structure of the article

The remainder of this article is structured as follows. [Section 2](#) surveys related work in industrial anomaly detection. [Section 3](#) specifies the proposed RBC-AD framework. [Section 4](#) details the experimental configuration and reports results on the TEP benchmark. [Section 5](#) presents comparative analysis with existing approaches. [Section 6](#) closes the paper.

2. Related work & background analysis

Fault and anomaly detection in the Tennessee Eastman Process (TEP) remain technically challenging because process variables are coupled, process dynamics are nonlinear, and fault effects may propagate with delay. Classical multivariate statistical monitoring methods, including PCA and distributed PCA variants, estimate dominant variance subspaces and detect deviations through SPE and T^2 statistics (Gao et al., 2022). These methods provide a linear variance-subspace formulation, but their linear and quasi-stationary assumptions can limit sensitivity to nonlinear faults and transient operating regimes. Quality-oriented Recursive-CPLS separates process-relevant and quality-relevant variation and supports recursive model updating; its decision layer, however, remains tied to linear latent structure and fixed decision rules (Hu et al., 2019). Graph-based predictable feature analysis represents multivariate temporal relations under correlated dynamics, although threshold stability under operating shifts remains unresolved (Fan et al., 2023).

Deep learning methods address nonlinear sequence representation in process monitoring. Unsupervised deep anomaly detection benchmarks on TEP report F1 and AUPRC for reconstruction, forecasting, generative, and hybrid model families, but deployment calibration and stream-level false-alarm-rate control are not standardised (Hartung et al., 2023). LSTM-attention and dual-attention autoencoder models capture temporal and feature dependencies; most reported evaluations, however, use sampled windows and fixed operating points (Zeng et al., 2024; Zhao et al., 2025). Real-time schemes target low-latency detection, and dynamic-loss VAE models address sensitivity under heterogeneous conditions, but both remain dependent on method-specific thresholding (Attouri et al., 2024; Vijai & P, 2025). Fuzzy neural fault-detection models and adversarial TEP benchmarks extend robustness analysis, although uncertainty-aware calibration and nominal deployment control are treated as separate issues (Adeli & Mazinan, 2020; Pozdnyakov et al., 2024).

Recent process-monitoring research has moved toward fusion-based and explainable learning structures. A hybrid causal-inference and neuro-fuzzy framework combines causal structure with adaptive neuro-fuzzy reasoning for advanced process monitoring, supporting interpretability while retaining model-dependent alarm selection (Ali et al., 2026). The DiGLPP–Kolmogorov–Arnold network framework develops fault representation for industrial chemical processes, although explicit conformal uncertainty and stream FAR control are not central components of its decision layer (Ali et al., 2026). Dynamic explainable fusion monitoring combines machine-learning fusion with contribution analysis for industrial and chemical processes, but deployment thresholds remain separate from finite-sample calibration (Ali et al., 2025). A

Table 1. Comparison of representative TEP fault and anomaly detection approaches.

Ref.	Method	Key Limitation
Gao et al. (2022)	PCA-based distributed monitoring	Linear modelling assumption and fixed thresholds
Hu et al. (2019)	Recursive-CPLS monitoring with model updating	Linear latent structure and limited uncertainty handling
Fan et al. (2023)	Graph-based predictable feature analysis	Threshold stability under operating shifts not guaranteed
Hartung et al. (2023)	Deep anomaly detection benchmark on TEP	Calibration and stream FAR control not standardised
(Zeng et al., 2024; Zhao et al., 2025)	Attention-based recurrent and autoencoder models	Sampled-window evaluation and fixed operating points
(Attouri et al., 2024; Vijai & P, 2025)	Real-time and VAE-based anomaly detection	Deployment thresholding remains method-dependent
Ali et al. (2026)	Causal-inference and neuro-fuzzy monitoring	Alarm calibration remains model-dependent
Ali et al. (2026)	DiGLPP–Kolmogorov–Arnold network fault detection	Conformal uncertainty and stream FAR control not addressed
Ali et al. (2025)	Dynamic explainable fusion monitoring	Explanation emphasised, with limited bounded alarm control
Ali et al. (2025)	Reconstruction-based Bayesian dynamic fusion	Three-state deployment decisioning not provided
RBC-AD	Probabilistic temporal model + frequency fusion + conformal calibration	FAR-controlled, uncertainty-aware, stream-evaluated

reconstruction-based Bayesian fusion framework models dynamic industrial systems through probabilistic reconstruction, yet its evaluation does not directly address three-state alarm decisioning under leakage-controlled calibration splits (Ali et al., 2025). This comparison indicates that recent fusion methods mainly address feature representation, diagnostic explanation, and model integration, whereas the alarm threshold is rarely formulated as an explicitly budgeted deployment parameter. For operator-facing monitoring, this distinction matters because high discrimination scores do not by themselves define acceptable alarm rates, uncertainty handling, or detection persistence in continuous streams. Table 1 summarises representative TEP fault and anomaly detection approaches. Existing methods address representation, fusion, diagnosis, and interpretability, but three deployment gaps remain: threshold selection is often fixed or implicitly tuned, uncertainty states are seldom formalised, and reported performance commonly emphasises window-level metrics without full-stream validation. RBC-AD targets these gaps through probabilistic temporal scoring, frequency-domain deviation cues, conformal calibration, FAR-controlled threshold selection, and continuous-stream evaluation.

3. Proposed methodology

This work implements a leakage-aware fault detection pipeline over Tennessee Eastman Process (TEP) multivariate time series, combining (i) windowed one-step-ahead probabilistic forecasting using a temporal convolution and causal self-attention model, (ii) a calibrated frequency-domain deviation score, (iii) robust score normalisation and convex score fusion, and (iv) conformal quantile thresholds with a three-state decision rule and a deployment operating point selected to satisfy an empirical false-alarm-rate (FAR) target on a normal score stream. Evaluation modules quantify detection delay, perform drift-triggered threshold recalibration, and compute feature-level root-cause rankings from per-sensor negative log-likelihood (NLL) contributions.

3.1. Leakage-aware preprocessing, run-level splitting, and window reservoir sampling

Each record contains metadata columns `simulationRun`, `sample`, and `faultNumber`, plus numeric sensor and actuator features identified by the prefix rule `xmeas_` or `xmv_`, with a numeric fallback. For each dataframe, metadata columns are enforced, feature values are coerced to numeric, and duplicate keys (`simulationRun`, `sample`) are removed while preserving stable ordering by (`simulationRun`, `sample`).

Missing values are imputed using per-feature medians computed on the normal `train_fit` split only. Let $x_{r,t,f}$ denote feature f at sample index t in run r . Let $\mathcal{D}_{\text{fit}}$ be the set of rows in the normal training data restricted to runs assigned to `train_fit`. Define

$$1_{\text{fin}}(x) = \begin{cases} 1, & x \text{ is finite,} \\ 0, & x \text{ is NaN or } \pm \infty. \end{cases} \quad (1)$$

In implementation, $\pm \infty$ values are replaced with NaN before imputation. The imputation median for feature f is

$$m_f = \text{median}(\{x_{r,t,f} : (r, t) \in \mathcal{D}_{\text{fit}}, 1_{\text{fin}}(x_{r,t,f}) = 1\}),$$

$$x_{r,t,f} \leftarrow \begin{cases} x_{r,t,f}, & 1_{\text{fin}}(x_{r,t,f}) = 1, \\ m_f, & 1_{\text{fin}}(x_{r,t,f}) = 0. \end{cases} \quad (2)$$

All feature columns are stored as `float32`. Normal runs are partitioned into four disjoint run sets $\mathcal{R}_{\text{train}}$, $\mathcal{R}_{\text{val}}$, $\mathcal{R}_{\text{cal_thr}}$, and $\mathcal{R}_{\text{cal_lam}}$ by shuffling unique run identifiers with a fixed random seed and applying the ratio tuple `NORMAL_RUN_SPLIT_TRAIN`. A feature standardiser is fit on cleaned normal `train_fit` data only. For feature vector $\mathbf{x} \in \mathbb{R}^F$, scaling applies

$$\tilde{\mathbf{x}} = (\mathbf{x} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}, \quad (3)$$

where $(\boldsymbol{\mu}, \boldsymbol{\sigma})$ are the per-feature mean and standard deviation estimated on `train_fit`, and $\oslash$ denotes elementwise division. A window extractor produces overlapping windows of length W with stride s . For a scaled run matrix $\tilde{\mathbf{X}}_r \in \mathbb{R}^{T_r \times F}$, windows are

$$\mathbf{X}_{r,i} = \tilde{\mathbf{X}}_r[t_i : t_i + W - 1, :] \in \mathbb{R}^{W \times F}, \quad t_i = 1 + (i - 1)s, \quad (4)$$

with

$$i = 1, \dots, \left\lfloor \frac{T_r - W}{s} \right\rfloor + 1.$$

To cap memory usage while preserving an unbiased sample of the window stream, normal windows are selected with reservoir sampling to a fixed capacity N per split. Let $\{\mathbf{X}_j\}_{j=1}^M$ be the stream of candidate windows. For $j > N$, sample $u \sim \text{Unif}\{0, \dots, j - 1\}$ and replace reservoir element u if $u < N$. Fault datasets are stored by run. For each fault trajectory, window labels use the known injection time `INJECTION_SAMPLE`. If a window start index is t_i , the binary label is

$$y_{r,i} = 1[t_i \geq t_{\text{inj}}], \quad t_{\text{inj}} = \text{INJECTION_SAMPLE}. \quad (5)$$

The fault sampler optionally filters to post-injection windows only. Algorithm 1 formalises schema enforcement, leakage-safe imputation and scaling, run-level splitting, and window reservoir sampling.

Algorithm 1. Leakage-aware preprocessing and window sampling.

Input : Raw multivariate time series $\{\mathbf{X}_r\}_{r=1}^R$ with metadata
Output: Window tensors $\mathcal{W}_{\text{train}}, \mathcal{W}_{\text{val}}, \mathcal{W}_{\text{cal_thr}}, \mathcal{W}_{\text{cal_lam}}$

- 1 Enforce schema columns (`simulationRun`, `sample`, `faultNumber`)
- 2 Identify numeric sensor features $\mathcal{F}$ using prefix and dtype rules
- 3 Replace $\pm \infty$ with NaN and coerce features to `float32`
- 4 Split normal runs into $\mathcal{R}_{\text{train}}, \mathcal{R}_{\text{val}}, \mathcal{R}_{\text{cal_thr}}, \mathcal{R}_{\text{cal_lam}}$
- 5 Compute per-feature medians $\{m_f\}$ on $\mathcal{R}_{\text{train}}$
- 6 Impute missing values using $\{m_f\}$
- 7 Fit standard scaler $(\boldsymbol{\mu}, \boldsymbol{\sigma})$ on $\mathcal{R}_{\text{train}}$
- 8 Standardize all features
- 9 **foreach** *run* *r* **do**
- 10 Extract overlapping windows of length W with stride s
- 11 Apply reservoir sampling to cap window count per split
- 12 **return** window reservoirs for each split

3.2. Probabilistic temporal forecaster and NLL training objective

The temporal forecaster maps each window $\mathbf{X} \in \mathbb{R}^{W \times F}$ to one-step-ahead heteroscedastic Gaussian parameters aligned to next-step targets. The backbone comprises a causal temporal convolutional network (TCN) followed by causal multi-head self-attention. The TCN uses three `TemporalBlock` modules with channel widths (96, 96, 96), kernel size $k = 5$, and dilations (1, 2, 4). Each block uses left padding and a `Chomp1d` operation to remove the padded suffix and enforce causality. After the TCN, a 1×1 convolution maps to attention width D (ATTN_DIM), yielding latent sequence $\mathbf{Z} \in \mathbb{R}^{B \times W \times D}$. Causal self-attention applies a boolean mask blocking access to future indices. For sequence length $T = W$, the mask is:

$$\mathbf{M}_{ij} = 1 [j > i], \quad i, j \in \{1, \dots, T\}. \quad (6)$$

The final latent state is discarded to enforce one-step-ahead prediction over $t \in \{1, \dots, W - 1\}$. Let $\mathbf{z}_t \in \mathbb{R}^D$ denote the attention output at time t . Linear heads output

$$\boldsymbol{\mu}_t = \mathbf{W}_\mu \mathbf{z}_t + \mathbf{b}_\mu \in \mathbb{R}^F, \quad \ell_t = \mathbf{W}_\ell \mathbf{z}_t + \mathbf{b}_\ell \in \mathbb{R}^F, \quad \ell_t \leftarrow \text{clip}(\ell_t, \ell_{\min}, \ell_{\max}). \quad (7)$$

Training minimises Gaussian negative log-likelihood (NLL) averaged over batch, time, and feature dimensions. For a single time step and feature, with target x and parameters (μ, ℓ) where $\ell = \log \sigma^2$, the per-datum NLL term is

$$\mathcal{L}_{\text{nll}}(x; \mu, \ell) = \frac{1}{2}(\ell + \exp(-\ell)(x - \mu)^2) + \frac{1}{2} \log(2\pi). \quad (8)$$

3.3. Dual-view anomaly scoring: temporal NLL, frequency deviation, robust normalisation, and fusion

Given a window $\mathbf{X} \in \mathbb{R}^{W \times F}$, the model returns $(\boldsymbol{\mu}_{1:W-1,:}, \ell_{1:W-1,:})$ aligned to the next-step targets $\mathbf{X}_{2:W,:}$. The temporal score is the mean NLL:

$$s_{\text{temp}}(\mathbf{X}) = \frac{1}{(W-1)F} \sum_{t=1}^{W-1} \sum_{f=1}^F \mathcal{L}_{\text{nll}}(X_{t+1,f}; \mu_{t,f}, \ell_{t,f}). \quad (9)$$

Per-feature contributions for interpretability are obtained by averaging the NLL over time only:

$$\mathbf{c}(\mathbf{X}) \in \mathbb{R}^F, \quad c_f(\mathbf{X}) = \frac{1}{W-1} \sum_{t=1}^{W-1} \mathcal{L}_{\text{nll}}(X_{t+1,f}; \mu_{t,f}, \ell_{t,f}). \quad (10)$$

A frequency score is computed from the real FFT magnitude along the temporal axis. For window tensor $\mathbf{X} \in \mathbb{R}^{W \times F}$, define the per-window magnitude spectrum $\mathbf{A} \in \mathbb{R}^{K \times F}$ where $K = \lfloor W/2 \rfloor + 1$:

$$\mathbf{A} = |\text{rFFT}(\mathbf{X})|. \quad (11)$$

Frequency bins are partitioned into B disjoint index sets $\{\mathcal{I}_b\}_{b=1}^B$ by an equal split of $\{1, \dots, K\}$. The band feature vector $\mathbf{b} \in \mathbb{R}^B$ is

$$b_b(\mathbf{X}) = \frac{1}{|\mathcal{I}_b| F} \sum_{k \in \mathcal{I}_b} \sum_{f=1}^F A_{k,f}. \quad (12)$$

Calibration fits per-band mean and standard deviation on `x_cal_thr`, producing $(\boldsymbol{\mu}_b, \boldsymbol{\sigma}_b)$. Two additional scalar frequency features are used. Half-spectrum flux compares magnitudes computed on the first and second halves of the window:

$$\text{flux}(\mathbf{X}) = \frac{1}{KF} \sum_{k=1}^K \sum_{f=1}^F \max(0, A_{k,f}^{(2)} - A_{k,f}^{(1)}), \quad (13)$$

where $A^{(1)}$ and $A^{(2)}$ are FFT magnitudes computed from $\mathbf{X}_{1:\lfloor W/2 \rfloor,:}$ and $\mathbf{X}_{\lfloor W/2 \rfloor+1:W,:}$ respectively. Spectral entropy computes $p_{k,f} = A_{k,f} / (\sum_{k'} A_{k',f} + \varepsilon)$ and then averages

$$\text{ent}(\mathbf{X}) = \frac{1}{F} \sum_{f=1}^F \left(- \sum_{k=1}^K p_{k,f} \log(p_{k,f} + \epsilon) \right), \quad \epsilon = \text{FREQ_EPS}. \quad (14)$$

Equations (13) and (14) are standardised using `CAL_THR` mean and standard deviation. The frequency score is a fixed convex combination:

$$s_{\text{freq}}(\mathbf{X}) = 0.55 \cdot \max_{b \in \{1, \dots, B\}} \left| \frac{b_b(\mathbf{X}) - \mu_b}{\sigma_b} \right| + 0.30 \cdot \left| \frac{\text{flux}(\mathbf{X}) - \mu_{\text{flux}}}{\sigma_{\text{flux}}} \right| + 0.15 \cdot \left| \frac{\text{ent}(\mathbf{X}) - \mu_{\text{ent}}}{\sigma_{\text{ent}}} \right|. \quad (15)$$

Temporal and frequency scores are normalised using median and median absolute deviation (MAD) fit on `cal_thr` scores. For a score stream $\{s_{ij}\}_{i=1}^n$, define

$$\begin{aligned} \text{med} &= \text{median}(\{s_{ij}\}), \quad \epsilon_{\text{MAD}} = 0.1, \\ \text{mad} &= \max(\text{median}(\{|s_{ij} - \text{med}|\}) + 10^{-6}, \epsilon_{\text{MAD}}), \quad z_i = \frac{s_i - \text{med}}{\text{mad}}. \end{aligned} \quad (16)$$

Let $z_{\text{temp}}(\mathbf{X})$ and $z_{\text{freq}}(\mathbf{X})$ be the robust-normalised temporal and frequency scores. The fused score is a convex mixture with parameter $\lambda \in [0, 1]$:

$$s_{\lambda}(\mathbf{X}) = \lambda z_{\text{temp}}(\mathbf{X}) + (1 - \lambda) z_{\text{freq}}(\mathbf{X}). \quad (17)$$

Optionally, a smoothing operator applies a moving average of width L to the fused score sequence:

$$\bar{s}_{\lambda}[t] = \frac{1}{L} \sum_{j=0}^{L-1} s_{\lambda}[t - j]. \quad (18)$$

When $L > 1$, smoothing is implemented via convolution in `same` mode. The fusion weight λ is selected by grid search over $\{0, 0.05, \dots, 1\}$. For each candidate λ , an uncertainty threshold is computed on `cal_thr` fused scores and evaluated on a disjoint tuning set formed by concatenating normal windows from `cal_lam` and post-injection fault windows sampled from the fault training runs. The selection objective is fixed *a priori*:

$$J(\lambda) = 1.10 \text{ F1}(\lambda) + 0.15 \text{Rec}(\lambda), \quad (19)$$

where predictions are obtained by thresholding the fused score at the threshold computed for the same λ . Coefficients are held fixed across experiments. Algorithm 2 summarises dual-view scoring, robust normalisation, and convex fusion.

Algorithm 2. Dual-view anomaly scoring and fusion.

Input : Window $\mathbf{X} \in \mathbb{R}^{W \times F}$

Output: Fused anomaly score $s_{\lambda}(\mathbf{X})$

- 1 Compute temporal NLL score $s_{\text{temp}}(\mathbf{X})$ using Eq. (9)
 - 2 Compute FFT magnitude spectrum of $\mathbf{X}$
 - 3 Extract band energies, spectral flux, and spectral entropy
 - 4 Standardize frequency features using calibration statistics
 - 5 Compute frequency score $s_{\text{freq}}(\mathbf{X})$ using Eq. (15)
 - 6 Apply robust normalization (median, MAD) to both scores using Eq. (16)
 - 7 Fuse scores: $s_{\lambda}(\mathbf{X}) = \lambda z_{\text{temp}}(\mathbf{X}) + (1 - \lambda) z_{\text{freq}}(\mathbf{X})$
 - 8 **return** fused score $s_{\lambda}(\mathbf{X})$
-

3.4. Conformal decisioning, FAR-controlled deployment threshold, and drift-triggered recalibration

Let $\{u_i\}_{i=1}^n$ be calibration scores computed on `cal_thr` using λ^* . Finite-sample corrected quantiles are computed via order statistics:

$$k = \min(\max(\lceil (n+1)q \rceil, 1), n), \quad \hat{Q}(q) = u_{(k)}, \quad (20)$$

where $u_{(k)}$ is the k th order statistic and $q \in (0, 1)$. The base uncertainty threshold is

$$\tau_u^{\text{base}} = \hat{Q}(1 - \alpha), \quad \alpha = \text{ALPHA}. \quad (21)$$

A stricter anomaly threshold is computed at $\beta = \alpha / \text{ANOM_BETA_DIV}$:

$$\tau_a^{\text{base}} = \hat{Q}(1 - \beta), \quad \beta = \alpha / \text{ANOM_BETA_DIV}. \quad (22)$$

For fused score s , the three-state prediction is

$$\hat{y}^{(3)}(s) = \begin{cases} 0, & s < \tau_u, \\ 1, & \tau_u \leq s < \tau_a, \\ 2, & s \geq \tau_a, \end{cases} \quad (23)$$

corresponding to normal, uncertain, and anomaly. Binary views are reported via $1[\hat{y}^{(3)} \geq 1]$ and $1[\hat{y}^{(3)} = 2]$. Deployment adjusts the uncertainty threshold to satisfy an empirical FAR target on a normal-only score stream derived from `cal_thr` runs under fixed detection settings. An optional rolling maximum aggregator of width K is

$$s^{\max}[t] = \max_{j=t-K+1, \dots, t} s[j]. \quad (24)$$

Given the stream $\{s^{\max}[t]\}_{t=1}^T$, the FAR operating point is selected as the smallest threshold meeting alarm-rate target ρ :

$$\tau_u^{\text{far}} = \min \left\{ \tau: \frac{1}{T} \sum_{t=1}^T 1[s^{\max}[t] \geq \tau] \leq \rho \right\}. \quad (25)$$

This FAR-based selection defines a deployment operating point under a fixed detector configuration and does not add a conformal guarantee beyond the quantile computation on calibration scores. Calibration scores used for quantiles are computed on overlapping windows and are not independent; exchangeability is not enforced. In this work, thresholds are treated as finite-sample quantiles computed on the calibration score set under fixed preprocessing, windowing, and scoring settings, and their empirical behaviour is reported under the evaluation protocols.

An optional drift mechanism updates the deployed uncertainty threshold along a test normal score stream using a two-sample Kolmogorov-Smirnov (KS) test on a sliding window of size `DRIFT_KS_WINDOW`. Let $\mathbf{s}_{\text{ref}}$ denote the reference score stream from `cal_thr` normals under detection settings, and let $\mathbf{s}_{\text{win}}$ be the most recent window of test-normal scores. A drift event triggers when

$$p_{\text{KS}}(\mathbf{s}_{\text{win}}, \mathbf{s}_{\text{ref}}) < p_0, \quad D_{\text{KS}}(\mathbf{s}_{\text{win}}, \mathbf{s}_{\text{ref}}) > d_0, \quad (26)$$

where (p_0, d_0) correspond to `DRIFT_KS_P_THRESHOLD` and `DRIFT_KS_D_THRESHOLD`. Scores are buffered for recalibration using only windows predicted as normal, implemented as $s[t] < \tau_u[t]$ when `DRIFT_ONLY_USE_PRED_NORMAL=True`. When a drift trigger occurs and the buffer size is at least `DRIFT_MIN_BUFFER_FOR_RECAL`, a new threshold is computed on buffered scores and blended into the current threshold using an exponential moving average (EMA), while restricting downward shifts:

$$\tilde{\tau}_u = \hat{Q}_{\text{buf}}(1 - \alpha), \quad \tilde{\tau}_u \leftarrow \max(\tilde{\tau}_u, \kappa \tau_u), \quad \tau_u \leftarrow (1 - \eta) \tau_u + \eta \tilde{\tau}_u, \quad (27)$$

where $\kappa = \text{DRIFT_THR_DROP_FLOOR}$ and $\eta = \text{DRIFT_EMA_BLEND}$. The mechanism is capped to `DRIFT_MAX_EVENTS` recalibration events.

3.5. Interpretability and evaluation: root-cause ranking, delay computation, and full-stream protocols

Root-cause analysis (RCA) uses per-feature NLL contribution vectors from Equation (10), standardised using calibration statistics computed on `cal_thr`. Contribution statistics are

$$\mu_f = \mathbb{E}[c_f], \quad \sigma_f = \sqrt{\text{Var}(c_f)} + 10^{-6}, \quad z_f = \text{clip}\left(\frac{c_f - \mu_f}{\sigma_f}, -8, 8\right). \quad (28)$$

Let $\mathbf{Z} \in \mathbb{R}^{n \times F}$ stack clipped standardised vectors. A sparse precision matrix $\Theta \in \mathbb{R}^{F \times F}$ is estimated by Graphical Lasso with regularisation `RCA_GLASSO_ALPHA`, yielding a sparse conditional-association proxy over contribution coordinates under nominal data. Graph edges are retained when $|\Theta_{ij}| \geq \text{RCA_GRAPH_MIN_ABS_W}$. At inference, a contribution vector $\mathbf{c}$ is transformed to standardised magnitudes $\mathbf{b} = |\mathbf{z}|$. A propagated score combines direct magnitudes with a single propagation step using the absolute precision matrix:

$$\mathbf{p} = \mathbf{b} + \gamma \mathbf{W} \mathbf{b}, \quad \mathbf{W} = |\Theta|, \quad \gamma = \text{RCA_PROPAGATION_GAMMA}, \quad (29)$$

with diagonal entries of $\mathbf{W}$ set to zero. Rankings are reported for top- K features under $\mathbf{b}$ and $\mathbf{p}$ with $K = \text{RCA_TOPK}$. This RCA layer is interpretive and does not affect scoring, calibration, threshold selection, or decision logic. Detection delay is evaluated on fault trajectories from the test fault dataset. Each trajectory produces a sequence of scored windows with detection stride `DETECT_STRIDE`, optional smoothing, and optional rolling maximum aggregation. Using the deployed uncertainty threshold τ_u^{far} , define a binary alarm stream $a[t] = 1[s[t] \geq \tau_u^{\text{far}}]$ aligned to window start samples $\{t_i\}$. Let k be the required number of consecutive alarms. The first index achieving k consecutive alarms is

$$i^* = \min \left\{ i: \sum_{j=0}^{k-1} a[i+j] \geq k \right\}. \quad (30)$$

Delay in samples is computed relative to the injection time, while enforcing post-injection detection:

$$\Delta = t_{i^*} - t_{\text{inj}}, \quad t_{\text{inj}} = \text{INJECTION_SAMPLE}. \quad (31)$$

The procedure returns no detection if no post-injection k -consecutive segment exists. Two operating points are reported, `DELAY_K_BALANCED` and `DELAY_K_RECALL`.

4. Simulation and performance evaluation

Experiments were conducted in Google Colab using Python, with PyTorch for model training and PyArrow for Parquet-backed access to Tennessee Eastman Process streams. The dataset represents TEP plant-wide process streams, where fault effects may appear through reactor dynamics, separation behaviour, recycle interactions, and closed-loop control responses. Fault-free and faulty trajectories (500 simulation runs per split) were processed by standardising metadata to `faultNumber`, `simulation_Run`, and `sample`, coercing non-numeric entries to numeric values, filtering invalid or missing sensor readings, and sorting each run by sample index to preserve temporal order. After preprocessing, 52 process variables (`xmeas_`, `xmv_`) were retained. Fault-free runs were partitioned at the run level into `train_fit` (325), `val_fit` (50), `cal_thr` (75), and `cal_lam` (50) to avoid leakage; standardisation parameters were fit using `train_fit` only. Windows with length $W = 100$ and stride 10 were generated by reservoir sampling, producing X_{train} (12000), X_{val} (2050), $X_{\text{cal_thr}}$ (3075), and X_{test} (11000; 3000 normal, 8000 post-injection fault). A causal dilated TCN with residual blocks and multi-head causal self-attention predicted Gaussian parameters and minimised Gaussian NLL using AdamW (lr 2×10^{-4} , weight decay 10^{-4}), gradient clipping (1.0), `ReduceLROnPlateau`, and early stopping (patience 10); the model contained 392264 parameters. Detection combined window-averaged NLL with an FFT-based frequency score normalised on `cal_thr`; $\lambda = 0.95$ was tuned and conformal quantiles defined thresholds at $\alpha = 0.05$, yielding sampled ROC AUC and PR AUC of 1.0, with FAR control at 0.05 and delay measured after injection at sample 160 for $k \in \{1, 3\}$.

4.1. Training evolution and performance convergence analysis

Training and validation represents the stable optimisation and convergence of the probabilistic forecaster used in RBC-AD. As shown in Figure 2, Gaussian negative log-likelihood (NLL) decreased from 1.4074 at epoch 1 to 1.1309 at epoch 50, and validation NLL decreased from 1.3217 to 1.1289; the curves remained

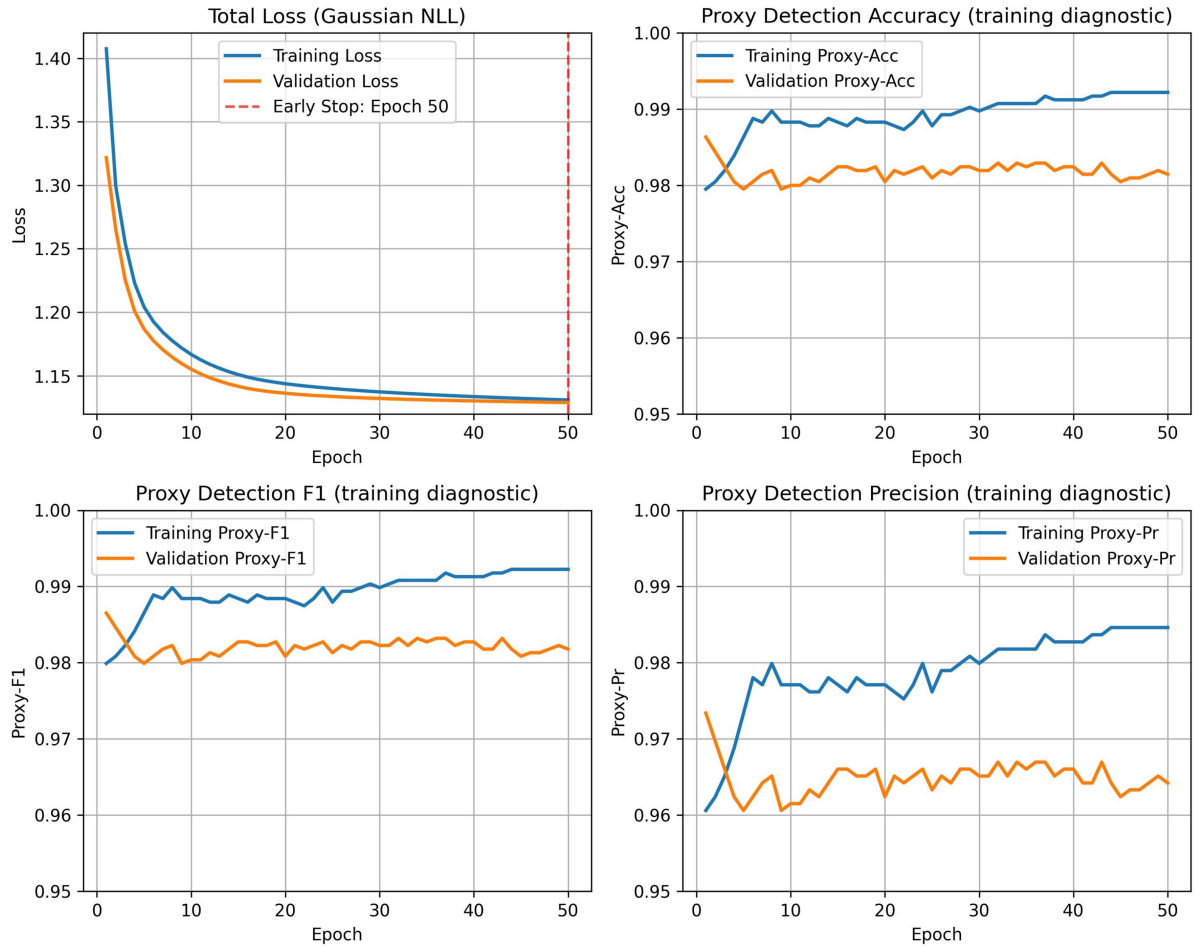


Figure 2. Epoch-wise training and validation Gaussian NLL with proxy detection accuracy, proxy F1-score, and proxy precision computed from thresholded NLL scores as training diagnostics; these proxy metrics are not used as supervised learning objectives.

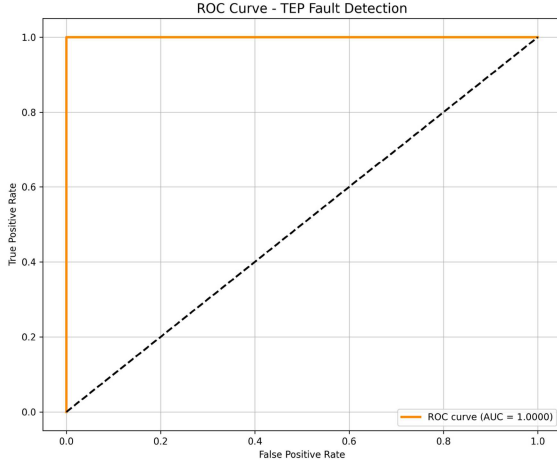
closely aligned without late-epoch divergence. Proxy detection metrics are reported as diagnostics computed from fixed-threshold NLL scores and were not used as optimisation targets; training minimises Gaussian NLL exclusively. Training proxy accuracy increased from 0.9795 to 0.9922, whereas validation proxy accuracy remained within 0.9795–0.9863 across the reported checkpoints. Training proxy precision increased from 0.9606 to 0.9846, while validation proxy precision remained within 0.9606–0.9734. Training proxy F1-score increased from 0.9799 to 0.9922, and validation proxy F1-score remained within 0.9799–0.9865. After approximately 15–20 epochs, metric trajectories showed limited variation, and the retained checkpoint at epoch 50 corresponds to the minimum validation loss (1.1289), consistent with convergence under AdamW with gradient clipping, ReduceLROnPlateau, and the specified early-stopping criterion.

4.2. Sampled-window test performance

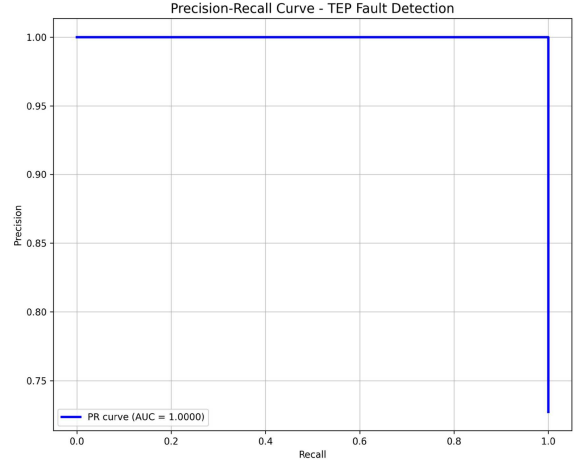
The sampled-window protocol assesses separability of the continuous combined score $s(\cdot)$ between fault-free operation and post-injection faulty operation. The test set comprises 11,000 windows, including 3,000 fault-free windows drawn from normal test runs and 8,000 windows sampled exclusively from the post-injection segment of faulty runs with window start ≥ 160 , using window length $W = 100$ and stride 10. Labels are assigned as $y = 0$ for fault-free windows and $y = 1$ for post-injection fault windows, and the pre-injection segment is excluded. This diagnostic does not characterise online detection on mixed trajectories and is complemented by FAR-controlled streaming and delay analyses (Section 4.3 and subsequent

Table 2. Sampled post-injection window test results under static conformal thresholds ($\theta_u = 0.1977$, $\theta_a = 0.3033$). Metrics are reported for the conservative ($s \geq \theta_u$) and strict ($s \geq \theta_a$) operating points; “Unc.” denotes the fraction of windows assigned to the *Uncertain* state, reported for all windows and stratified by true normal/fault.

Setting	Acc.	Prec.	F_1	Unc. (all)	Unc. (N/F)
Conservative ($s \geq \theta_u$)	0.9985	0.9979	0.9989	0.001364	0.005000/0.000000
Strict ($s \geq \theta_a$)	0.9998	0.9998	0.9999	0.001364	0.005000/0.000000



(a) ROC on sampled windows.



(b) PR on sampled windows.

Figure 3. ROC and PR curves computed from continuous combined scores $s(\cdot)$ under the sampled post-injection window protocol (3,000 fault-free and 8,000 post-injection fault windows). ROC-AUC and PR-AUC equal 1.000 under this separability diagnostic and should not be interpreted as full-stream mixed-trajectory performance.

sections). The fusion weight combining temporal and frequency scores was tuned without leakage using disjoint normal and post-injection fault windows, yielding $\lambda = 0.95$ with tuning $F_1 = 0.9867$ and precision 0.9737. Static conformal thresholds computed on the held-out calibration-normal split at $\alpha = 0.05$ with finite-sample correction yielded $\theta_u = 0.1977$ and $\theta_a = 0.3033$. Under the conservative operating point ($s \geq \theta_u$), accuracy was 0.9985, precision 0.9979, and F_1 was 0.9989; under the strict operating point ($s \geq \theta_a$), accuracy was 0.9998, precision 0.9998, and F_1 was 0.9999 as shown by Table 2. The three-state decision assigned 15 windows to *Uncertain* (0.001364 of all windows), with all assignments on fault-free windows (0.0050 of true normal windows) and none on post-injection fault windows. Continuous-score separability under this post-injection-only protocol yields ROC-AUC = 1.000 and PR-AUC = 1.000, as shown by the ROC curve in Figure 3(a) and the PR curve in Figure 3(b). These values quantify separability under the diagnostic sampling procedure and do not imply equivalent performance for full-stream trajectories containing both pre-injection and post-injection segments.

4.3. FAR-controlled operating point on test windows

A deployment operating point is defined by selecting an uncertainty threshold $\theta_{u,\text{deploy}}$ to satisfy an *empirical* normal alarm-rate target of 0.05 on a held-out calibration-normal score stream constructed from the CAL_THR normal runs under the specified detection windowing configuration. This procedure is restricted to operating-point selection and does not alter model fitting, fusion-weight tuning, or conformal calibration. Using 6,075 CAL_THR stream windows, the selected threshold is $\theta_{u,\text{deploy}} = 0.2546$, which yields an alarm rate of 0.04988 on the selection stream, while the anomaly threshold remains fixed at $\theta_a = 0.3033$. When applied to the sampled test windows, the FAR-selected threshold produces 2 *Uncertain* assignments (0.000182 overall), all on fault-free windows (0.000667 of true normal windows) and none on post-injection fault windows. Under $\theta_{u,\text{deploy}}$, the conservative operating point ($s \geq \theta_{u,\text{deploy}}$) yields accuracy

Table 3. Sampled-window test results under FAR-controlled deployment threshold ($\theta_{u,\text{deploy}} = 0.2546$, $\theta_a = 0.3033$). “Unc.” denotes the fraction of windows assigned to the *Uncertain* state, reported for all windows and stratified by true normal/fault.

Setting	Acc.	Prec.	F_1	Unc. (all)	Unc. (N/F)
Conservative ($s \geq \theta_{u,\text{deploy}}$)	0.9996	0.9995	0.9998	0.000182	0.000667/0.000000
Strict ($s \geq \theta_a$)	0.9998	0.9998	0.9999	0.000182	0.000667/0.000000

Table 4. Full-stream test performance under static thresholds ($\theta_u = 0.1977$, $\theta_a = 0.3033$), $W = 100$, stride 10, smoothing window 3. “Unc.” denotes the overall uncertain-rate under the three-state decision rule.

Protocol	Setting	Acc.	Prec.	Rec.	F_1	Unc. (all)
Post_only	Conservative ($\geq \theta_u$)	0.8643	0.9974	0.8582	0.9226	0.0100
Post_only	Strict ($\geq \theta_a$)	0.8574	0.9994	0.8492	0.9182	0.0100
Mixed	Conservative ($\geq \theta_u$)	0.8133	0.8971	0.8582	0.8772	0.0133
Mixed	Strict ($\geq \theta_a$)	0.8126	0.9039	0.8492	0.8757	0.0133

0.9996, precision 0.9995, and F_1 0.9998, while the strict operating point ($s \geq \theta_a$) yields accuracy 0.9998, precision 0.9998, and F_1 0.9999 as shown by Table 3. ROC-AUC and PR-AUC are invariant to operating-point selection because they are computed from the continuous score $s(\cdot)$ rather than thresholded decisions.

4.4. Full-stream test evaluation

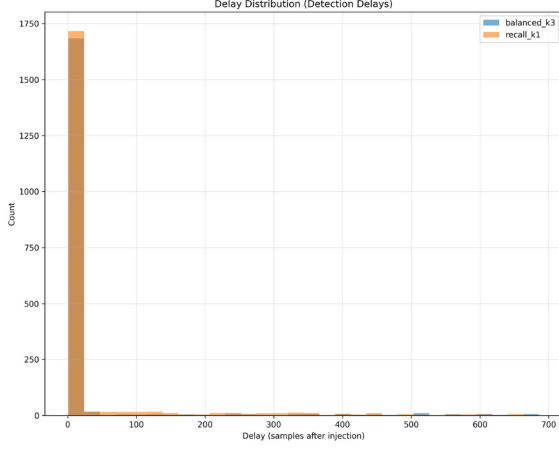
Full-stream performance is evaluated using static conformal thresholds derived from calibration-normal data, with $\theta_u = 0.1977$, $\theta_a = 0.3033$, window length $W = 100$, stride 10, and score-smoothing window 3. Results are reported under two labelling protocols: **post_only**, which restricts faulty trajectories to post-injection windows and pairs them with normal trajectories, and **mixed**, which includes both pre-injection and post-injection windows from faulty trajectories as shown by Table 4. Under **post_only** (753, 500 windows), the conservative operating point ($s \geq \theta_u$) yields accuracy 0.8643, precision 0.9974, recall 0.8582, and F_1 0.9226, with specificity 0.9639 (FPR 0.0361) and FNR 0.1418; the strict operating point ($s \geq \theta_a$) yields accuracy 0.8574, precision 0.9994, recall 0.8492, and F_1 0.9182, with specificity 0.9911 (FPR 0.00894) and FNR 0.1508. The three-state decision yields an uncertain rate of 0.0100 overall, with 0.0271 on true normal windows and 0.00895 on true fault windows. Under **mixed** (913, 500 windows), the conservative operating point yields accuracy 0.8133, precision 0.8971, recall 0.8582, and F_1 0.8772, with specificity 0.6567 (FPR 0.3433) and FNR 0.1418; the strict operating point yields accuracy 0.8126, precision 0.9039, recall 0.8492, and F_1 0.8757, with specificity 0.6850 (FPR 0.3150) and FNR 0.1508. The uncertain rate under the mixed protocol is 0.0133 overall, with 0.0283 on true normal windows and 0.00895 on true fault windows. The increase in false positives under **mixed** is primarily attributable to scoring pre-injection segments of faulty runs, where early transient behaviour overlaps with normal operating dynamics under thresholds fixed from calibration-normal scores.

4.5. Detection delay and fault-wise miss-rate diagnostics

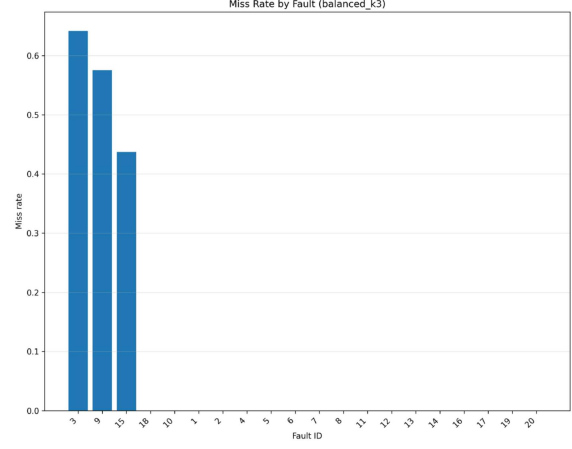
Detection delay is evaluated on 2, 000 test fault trajectories using the FAR-controlled deployment threshold $\theta_{u,\text{deploy}} = 0.2546$ with detection stride 5, smoothing window 1, and score aggregation window 1. Delay is measured in samples after the injection point using the first post-injection window that satisfies a k -consecutive alarm rule. The balanced configuration ($k = 3$) yields median delay 1 and P95 215, with 178 trajectories never detected; the recall-oriented configuration ($k = 1$) yields median delay 1 and P95 286, with 45 trajectories never detected as shown in Table 5. Figure 4(a) shows that both configurations concentrate a large proportion of detections at minimal delay, with a right-skewed distribution and a long tail corresponding to late detections. The $k = 3$ configuration exhibits a comparatively shorter upper tail than $k = 1$, consistent with the lower P95 value, while $k = 1$ produces fewer never-detected trajectories at the expense of heavier tail behaviour. A trajectory is classified as never detected when no post-injection window satisfies the k -consecutive alarm criterion. Fault-wise miss rate is defined as the fraction of trajectories for a given fault ID that are never detected. Under the balanced configuration, nonzero miss rates are concentrated on faults 3, 9, and 15, as illustrated in Figure 4(b), while other faults have near-zero miss rates under the

Table 5. Detection delay statistics on 2, 000 test fault trajectories under $\theta_{u,\text{deploy}} = 0.2546$, detection stride 5, smoothing window 1, and score aggregation window 1.

Setting	k	Median	P95	Never detected
Balanced	3	1	215	178
Recall-oriented	1	1	286	45



(a) Delay distribution for $k \in \{1, 3\}$.



(b) Fault-wise miss rate for the balanced configuration ($k = 3$).

Figure 4. Delay and fault-wise miss-rate diagnostics under $\theta_{u,\text{deploy}} = 0.2546$ with detection stride 5.

evaluated setting. This behaviour is consistent with the physical characteristics of these TEP faults. Faults 3 and 9 correspond to changes in the D-feed temperature, where the resulting process deviations can be weakly expressed in the measured variables because closed-loop control compensates part of the disturbance. Fault 15 corresponds to condenser cooling water valve sticking, which can produce intermittent or slowly developing effects rather than a sharp post-injection deviation. These fault mechanisms reduce the persistence of the anomaly score and make it harder to satisfy the balanced $k = 3$ consecutive-alarm rule. Therefore, the higher miss rates for faults 3, 9, and 15 are attributed to weak observability, controller compensation, and non-persistent score excursions rather than random model failure.

4.6. Normal-stream alarm-rate reporting and drift-triggered threshold adaptation

Normal-stream deployment behaviour is evaluated under the detection configuration (stride 5, smoothing window 1, score aggregation window 1) using $\theta_{u,\text{deploy}} = 0.2546$ selected on the CAL_THR normal stream to meet an empirical normal alarm-rate target of 0.05. On the held-out test normal stream, the achieved alarm rate is 0.03976, which is below the selection-stream alarm rate 0.04988 and below the FAR target. A KS-test drift monitor is applied to the normal score stream to trigger updates of the uncertainty threshold $\theta_u(t)$ while holding θ_a fixed; Figure 5 depicts the score stream, detected drift events, and the resulting threshold timeline used for optional adaptation under fault-free operation. Drift detection and threshold updates use only normal-stream scores under the configured KS windows and decision thresholds, without access to test labels.

4.7. Root-cause dependency graph analysis

Root-cause attribution is supported by a sparse dependency graph estimated from *per-feature Gaussian NLL contribution* statistics computed on the CAL_THR normal split and held fixed in all subsequent experiments to avoid test leakage. Per-feature contributions are defined as the time-average of Gaussian negative log-likelihood terms for each variable within a window, producing a feature-wise NLL contribution vector.

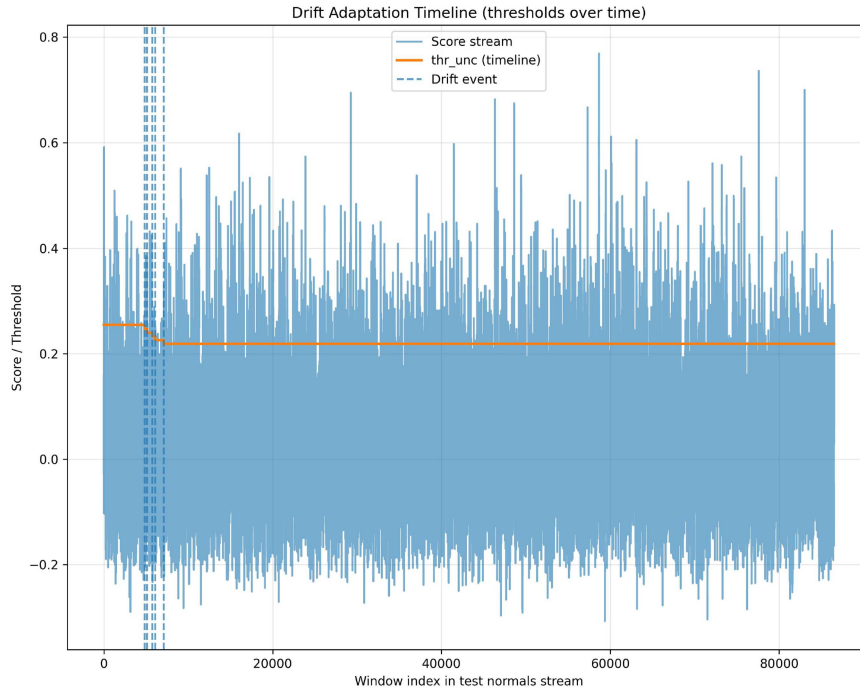


Figure 5. Normal-stream drift adaptation timeline showing the score stream, drift events, and the time-varying uncertainty threshold $\theta_u(t)$ under KS-test triggering; θ_a is held fixed.

Edges represent conditional dependencies under nominal operation through absolute precision-matrix coefficients fitted on standardised NLL contributions using GraphicalLasso. Low-magnitude edges are removed for visualisation; the reported graph retains 591 undirected edges as shown in Figure 6, and node size scales with weighted degree to emphasise variables exhibiting high aggregate connectivity across x_{meas}/x_{mv} groups. Root-cause reporting is performed post hoc on 500 test fault trajectories by selecting, for each trajectory, the first post-injection window assigned *Anomaly* (or *Uncertain* if no anomaly occurs) under the FAR-controlled deployment threshold $\theta_{u,deploy} = 0.2546$ with fixed anomaly threshold $\theta_a = 0.3033$. Within each selected window, variables are ranked by standardised NLL contribution and interpreted using immediate graph neighbourhoods to identify co-deviating variable sets consistent with the learned nominal dependency structure. The dependency graph does not participate in scoring, threshold selection, calibration, or decision logic and serves exclusively as an interpretability layer for organising NLL-based per-feature attributions.

5. Comparison against existing approaches

Published studies on Tennessee Eastman Process (TEP) monitoring span latent-variable statistical monitoring, deep unsupervised anomaly detection, supervised fault diagnosis, and robustness-oriented evaluation as shown by Table 6. Recursive-CPLS (RCPLS) addresses online model updating and reports monitoring measures including fault detection rate (FDR), false alarm rate (FAR), and detection delay, without ROC-AUC or AUPRC (Hu et al., 2019). Graph-based predictable feature analysis (GPFA) reports per-fault and averaged FDR/FAR under T^2 and SPE statistics and benchmarks against PCA and dynamic variants, emphasising behaviour under dynamic regimes (Fan et al., 2023). Unsupervised deep anomaly detection benchmarks on TEP report F1 and AUPRC across reconstruction, forecasting, generative, and hybrid model families, supporting ranking under class imbalance (Hartung et al., 2023). A distributed PCA monitoring study reports fault detection rates for selected TEP faults under PCA variants and related statistics, without ROC-AUC or AUPRC (Gao et al., 2022).

Supervised LSTM-attention fault diagnosis reports training and validation accuracy together with macro and micro ROC-AUC for multiclass fault identification on enriched TEP, which is not directly comparable to

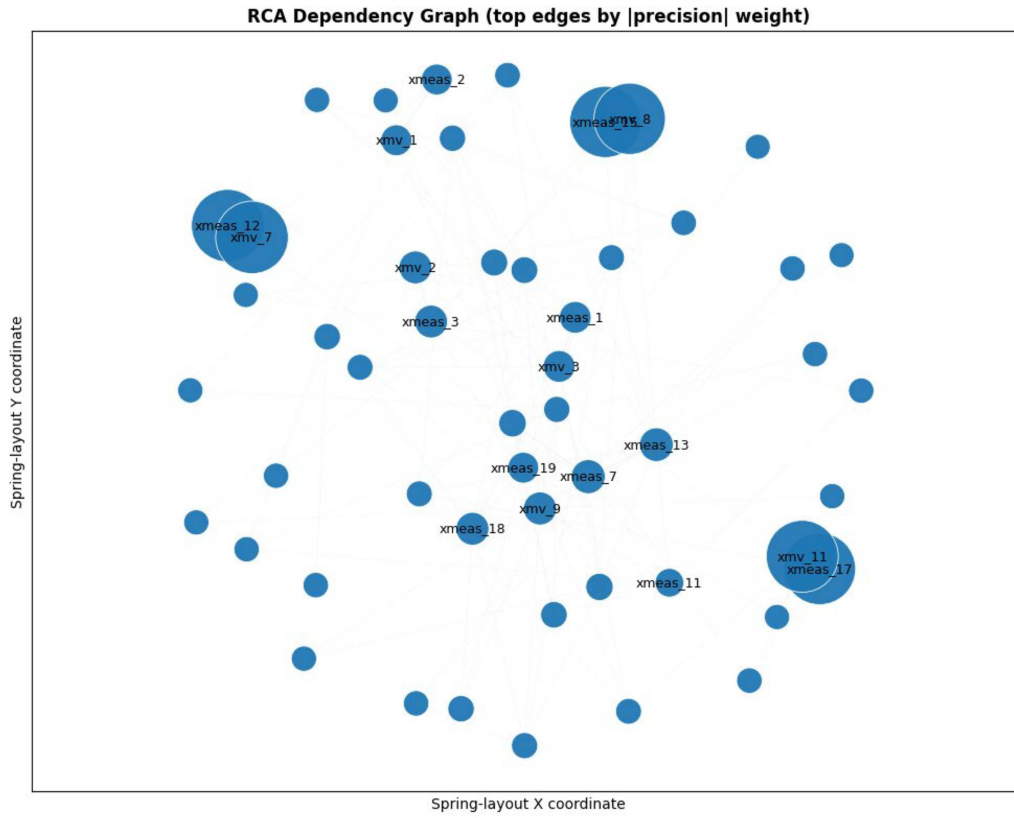


Figure 6. Dependency graph estimated from CAL_THR normal per-feature Gaussian NLL contributions; nodes denote process variables, edges encode conditional dependencies under nominal operation weighted by absolute precision-matrix magnitudes, node size denotes weighted degree, and edges below a fixed weight threshold are pruned (591 retained undirected edges).

Table 6. Comparison of existing models with the proposed RBC-AD framework.

Ref.	Method	Dataset	Metrics	Best reported values
Hu et al. (2019)	RCPLS (online CPLS update)	TEP	FDR, FAR, delay	FDR (RCPLS): $T_{cnew}^2 = 40.54\text{--}98.55$, $T_{xnew}^2 = 94.39\text{--}99.79$; FAR (RCPLS): $T_{cnew}^2 = 3.53\text{--}18.29$, $T_{xnew}^2 = 11.02\text{--}99.79$; delay (RCPLS): $T_{cnew}^2 = 0\text{--}94$, $T_{xnew}^2 = 0\text{--}19$
Hartung et al. (2023)	Unsupervised AD benchmark	TEP	F1, AUPRC	Best F1: BeatGAN = 0.9699; best AUPRC: TCN-S2S-AE = 0.9914
Fan et al. (2023)	GPFA (dynamic monitoring)	TEP	FDR, FAR	Avg (selected params): T^2 FDR = 80.20, FAR = 2.71; SPE FDR = 79.16, FAR = 2.38
Gao et al. (2022)	NewPCA (distributed PCA)	TEP	Fault detection rate	Selected faults: $ST^2 = 0.0\text{--}99.3\%$, $SSPE = 55.4\text{--}99.2\%$; example summary: $ST^2 = 75.5\%$, $SSPE = 77.5\%$
Zhao et al. (2025)	LSTM-Attention (supervised diagnosis)	Enriched TEP	Acc., ROC-AUC	Training/validation accuracy reported as $\geq 95.43\%$ after early epochs; macro ROC-AUC = 0.98, micro ROC-AUC = 0.98 (17 faults)
Vijai and P (2025)	BiLSTM-VAE + dynamic loss	TEP	Acc., F1	Mean Acc = 0.92; Mean F1 = 0.85
Pozdnyakov et al. (2024)	Adversarial benchmark (attack/defence)	TEP	Acc.	Nominal examples reported: TCN = 0.90; Quantisation ($n = 8$) = 0.89; AE = 0.75
Adeli and Mazinan (2020)	Fuzzy NN + fault-tolerant control	TEP	Multi-class accuracy	Overall accuracy $\approx 90.1\%$ (confusion-matrix evaluation)
Ours	RBC-AD (conformal + FAR control)	TEP	ROC-AUC, PR-AUC; stream F1; delay	Sampled windows: ROC-AUC = 1.0000, PR-AUC = 1.0000; full-stream F1 (post-fault only/mixed) and delay statistics reported in Tables 4 and 5.

unsupervised anomaly scoring (Zhao et al., 2025). A VAE-based method with dynamic loss weighting reports mean accuracy and mean F1 for multiclass anomaly classification on TEP (Vijai & P, 2025). Adversarial benchmarking evaluates accuracy under attack and defence settings for TEP fault detection models, focusing on robustness rather than nominal separability (Pozdnyakov et al., 2024). A fuzzy neural-

network fault detection and fault-tolerant control study reports multiclass classification accuracy derived from confusion-matrix evaluation and control-oriented behaviour, without ROC-AUC or AUPRC (Adeli & Mazinan, 2020). RBC-AD reports sampled-window separability metrics (ROC-AUC, PR-AUC) and full-stream operational metrics (F1, uncertainty rate, detection delay), and comparisons are limited to metric families explicitly reported without introducing unreported measures.

6. Conclusion

This study introduced Risk-Budgeted Conformal Anomaly Detection (RBC-AD) for industrial fault monitoring on the Tennessee Eastman Process. The approach couples a causal probabilistic forecaster with frequency-domain deviation features and uses conformal calibration to generate a three-state decision output (normal, uncertain, anomaly) at a specified miscoverage level. Model fitting, fusion-weight tuning, and calibration are separated at the run level to avoid leakage, and deployment thresholds are selected using normal-only calibration streams.

Under the sampled-window protocol restricted to post-injection fault segments, the continuous combined score attained ROC-AUC and PR-AUC of 1.000, reflecting complete separability under that diagnostic protocol. Full-stream evaluation provides a deployment-relevant assessment: under static thresholds, the conservative operating point yielded $F_1 = 0.9226$ on post_only trajectories and $F_1 = 0.8772$ on mixed trajectories, with uncertainty rates near 1%. A FAR-controlled deployment threshold selected to satisfy an empirical normal alarm-rate target generalised conservatively on the held-out normal stream. Detection-delay statistics and fault-wise miss rates further quantify timeliness and failure cases associated with difficult fault modes.

Future trends in industrial anomaly detection are expected to move toward adaptive calibration, process-aware model updating, uncertainty-aware alarm management, and operator-facing explanation systems. In this direction, future work should refine drift-triggered recalibration so that thresholds can adapt to gradual distributional changes without increasing false alarms. Further work should also improve graph-based root-cause attribution by integrating physical process knowledge and plant connectivity information with learned dependency structures.

For practical deployment, we recommend validating RBC-AD under longer continuous streams, multiple operating modes, and plant-specific alarm constraints before online use. Faults with weak observability or intermittent behaviour should be handled with fault-specific delay analysis, adaptive consecutive-alarm rules, and process-informed feature interpretation. Future studies may also evaluate the framework on additional industrial benchmarks and real plant data to assess generalisation, computational cost, and operator usability under realistic monitoring conditions.

Author contributions

CRedit: **Muhammad Mudasir:** Conceptualization, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft; **Yousef Asiri:** Investigation, Resources, Supervision, Validation, Writing – review & editing; **Iqra Ameer:** Conceptualization, Methodology, Validation, Visualization, Writing – review & editing; **Mana Saleh Al Reshan:** Resources, Supervision, Validation, Visualization, Writing – review & editing; **Hamad Almansour:** Investigation, Software, Validation, Writing – review & editing; **Kamran Ahmad Awan:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – review & editing; **Asadullah Shaikh:** Methodology, Supervision, Validation, Visualization, Writing – review & editing.

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Data availability statement

The data supporting the findings of this study are publicly available in the Harvard Dataverse repository as *Additional Tennessee Eastman Process Simulation Data for Anomaly Detection Evaluation* at <https://doi.org/10.7910/DVN/6C3JR1doi:10.7910/DVN/6C3JR1>. This study used the Tennessee Eastman Process (TEP) fault-free training, fault-free testing, faulty training,

and faulty testing files, provided as `TEP_FaultFree_Training.RData`, `TEP_FaultFree_Testing.RData`, `TEP_Faulty_Training.RData`, and `TEP_Faulty_Testing.RData`. The same dataset files are also available through the Kaggle repository titled *Tennessee Eastman Process Simulation Dataset* at <https://www.kaggle.com/datasets/averkij/tennessee-eastman-process-simulation-dataset>: Tennessee Eastman Process Simulation Dataset. The datasets were last accessed on 26 January 2026. The processed experimental outputs, trained model artifacts, configuration files, figures, and evaluation reports generated during this study are available from the corresponding author upon reasonable request.

List of acronyms

AUC	Area Under the Curve
AUPRC	Area Under the Precision-Recall Curve
CAL_LAM	Calibration Split for Fusion-Weight Tuning
CAL_THR	Calibration Split for Threshold Estimation
CPLS	Concurrent Projection to Latent Structures
DL	Deep Learning
EMA	Exponential Moving Average
FAR	False Alarm Rate
FDR	Fault Detection Rate
FFT	Fast Fourier Transform
FNR	False Negative Rate
FPR	False Positive Rate
FTC	Fault-Tolerant Control
GPFA	Graph-Based Predictable Feature Analysis
IIoT	Industrial Internet of Things
IoT	Internet of Things
KS	Kolmogorov-Smirnov
LSTM	Long Short-Term Memory
MAD	Median Absolute Deviation
ML	Machine Learning
NLL	Negative Log-Likelihood
NN	Neural Network
PCA	Principal Component Analysis
PR-AUC	Precision-Recall Area Under the Curve
RBC-AD	Risk-Budgeted Conformal Anomaly Detection
RCA	Root-Cause Attribution
RCPLS	Recursive Concurrent Projection to Latent Structures
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SPE	Squared Prediction Error
TCN	Temporal Convolutional Network
TEP	Tennessee Eastman Process
VAE	Variational Autoencoder

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